

## 1. INTRODUCTION

### 1.1 Project Overview

This project focuses on analyzing housing market trends using Tableau for visualization and Flask for web integration. The housing dataset contains details such as sale price, renovation year, number of bedrooms, bathrooms, floors, basement area, and house age.

The objective of this project is to transform raw housing data into interactive visual insights through dashboards and storyboards. The final dashboard and story are embedded into a web application using Flask.

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### 1.2 Purpose

- To analyze housing sale price trends
- To study renovation impact on sales
- To understand house age distribution
- To create interactive dashboard and story
- To integrate Tableau dashboard with Flask UI

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## 2. IDEATION PHASE

### 2.1 Problem Statement

Understanding housing market data is difficult when presented in raw tabular format. Buyers, sellers, and investors require clear visual insights to analyze:

- Sale price trends
- Renovation impact
- House age distribution
- Effect of bedrooms, bathrooms, and floors

Therefore, a visualization-based solution is required.

## Customer Problem Statement Template

I am

I'm trying to

But

Because

Which makes me feel

I am

I'm trying to

But

Because

Which makes me feel

A home buyer / real estate analyst who wants to understand housing market trends.

Analyze house prices and property features to make better buying decisions.

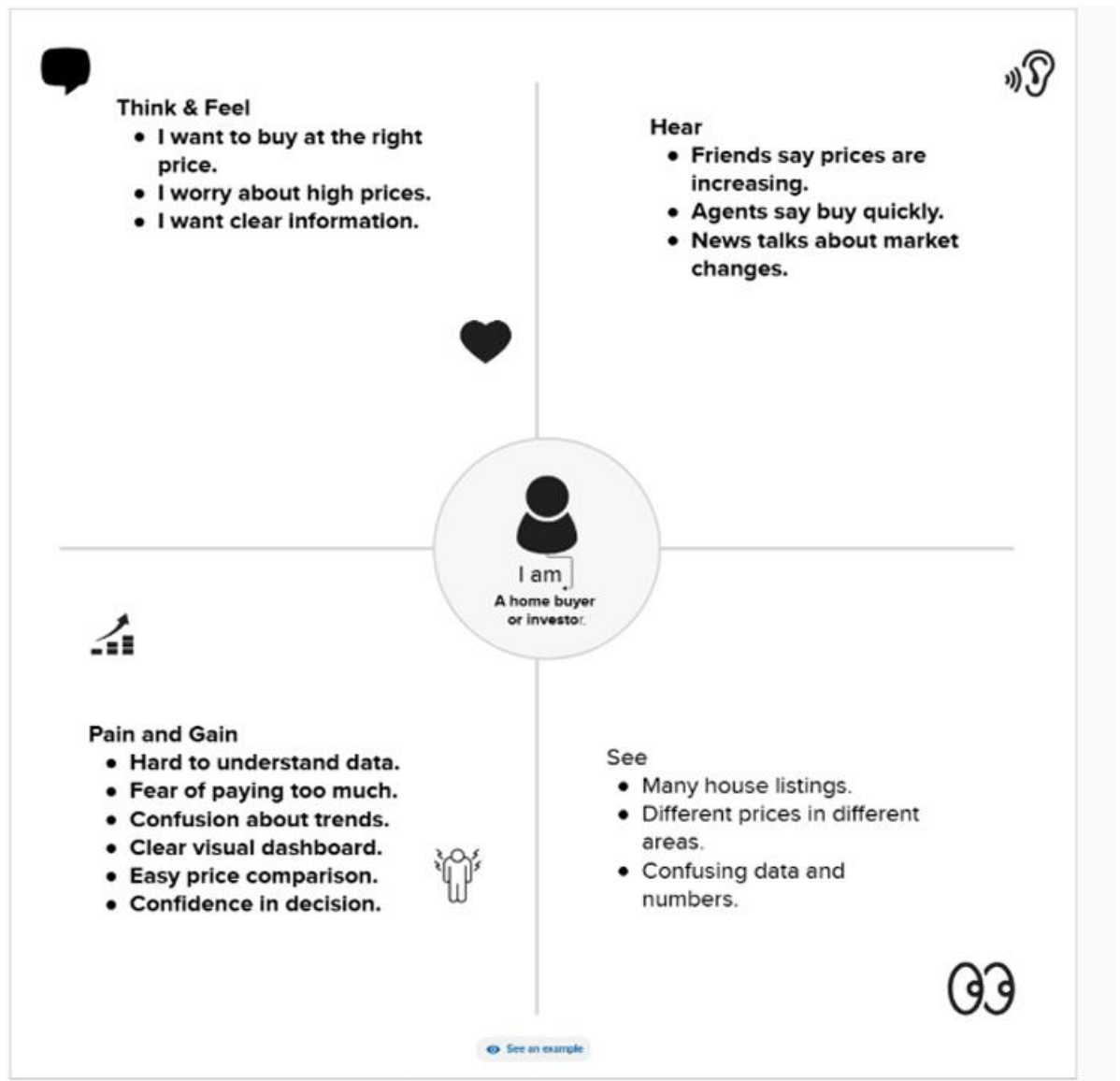
I find it difficult to understand raw housing data and price variations.

**The data is complex and not presented in a clear visual format.**

Confused and unsure about making the right investment decision.

## 2.2 Empathy Map Canvas

Empathy Map Canvas.



## 2.3 Brainstorming

Ideas discussed:

- Create sales overview visualization
- Show average sale price
- Show renovation impact
- Analyze house age
- Build interactive dashboard
- Integrate with Flask

Final Idea:

👉 Tableau Dashboard + Story + Flask Integration

### 3. REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

|  1. Entice<br>(Awareness)                                                                               |  2. Enter<br>(Access Website)                 |  3. Engage<br>(Explore Dashboard)       |  4. Exit<br>(Decision Making) |  5. Extend<br>(After Usage) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|  <b>Steps</b> <ul style="list-style-type: none"><li>• User wants to understand housing prices</li></ul> | <ul style="list-style-type: none"><li>• User wants to understand housing prices</li><li>• Searches for market trends</li></ul> | <ul style="list-style-type: none"><li>• Opens Flask web application</li><li>• Views embedded Tableau dashboard</li></ul> | <ul style="list-style-type: none"><li>• Analyzes trends.</li><li>• Makes buying/ investment decision</li></ul> | <ul style="list-style-type: none"><li>• Shares dashboard</li><li>• Revisits for future analysis</li></ul>      |
|  <b>Goal</b>                                                                                           | <ul style="list-style-type: none"><li>• Get clear price insights</li></ul>                                                     | <ul style="list-style-type: none"><li>• Explore housing data</li></ul>                                                   | <ul style="list-style-type: none"><li>• Explore housing data</li></ul>                                         | <ul style="list-style-type: none"><li>• Track market changes</li></ul>                                         |
|  <b>Pain</b><br>Pain (Challenges)                                                                     | <ul style="list-style-type: none"><li>• Raw data is confusing</li></ul>                                                        | <ul style="list-style-type: none"><li>• Too many charts may confuse</li></ul>                                            | <ul style="list-style-type: none"><li>• Clean layout and clear headings</li></ul>                              | <ul style="list-style-type: none"><li>• Add chart layout and headings</li></ul>                                |
|  <b>Opportunity</b>                                                                                   | <ul style="list-style-type: none"><li>• Provide simple dashboard homepage</li></ul>                                            | <ul style="list-style-type: none"><li>• Clean layout and clear headings</li></ul>                                        | <ul style="list-style-type: none"><li>• Add chart descriptions and tooltips</li></ul>                          | <ul style="list-style-type: none"><li>• Update dataset regularly</li></ul>                                     |
|  <b>Opportunity</b>                                                                                   | <ul style="list-style-type: none"><li>• Provide simple dashboard homepage</li></ul>                                            | <ul style="list-style-type: none"><li>• Clean layout and clear headings</li></ul>                                        | <ul style="list-style-type: none"><li>• Add chart descriptions and tooltips</li></ul>                          | <ul style="list-style-type: none"><li>• Add download/share feature</li></ul>                                   |

### 3.2 Solution Requirement

#### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)           |
|-------|-------------------------------|----------------------------------------------|
| FR-1  | User Registration             | Registration through Form (Email & Password) |
|       |                               | Registration through Gmail                   |
|       |                               | Registration through LinkedIn                |
| FR-2  | User Confirmation             | Confirmation via Email                       |
|       |                               | Confirmation via OTP                         |
|       |                               | Account activation after verification        |
| FR-3  | User Login & Authentication   | Login using Email & Password                 |
|       |                               | Login using Gmail                            |
|       |                               | Forgot Password option                       |
|       |                               | Logout functionality                         |
| FR-4  | Dashboard Access              | Access dashboard after login                 |
|       |                               | View housing price trends                    |
|       |                               | View renovation impact analysis              |
|       |                               | View sales distribution charts               |
| FR-5  | Data Filtering                | Filter by bedrooms                           |
|       |                               | Filter by location                           |
|       |                               | Filter by renovation status                  |
|       |                               | Filter by price range                        |
| FR-6  | Data Visualization            | Interactive Tableau dashboard embedding      |
|       |                               | Real-time chart updates                      |
| FR-7  | Admin Management              | Upload housing dataset                       |
|       |                               | Manage registered users                      |
|       |                               | Update/Delete dataset                        |

#### Non-functional Requirements:

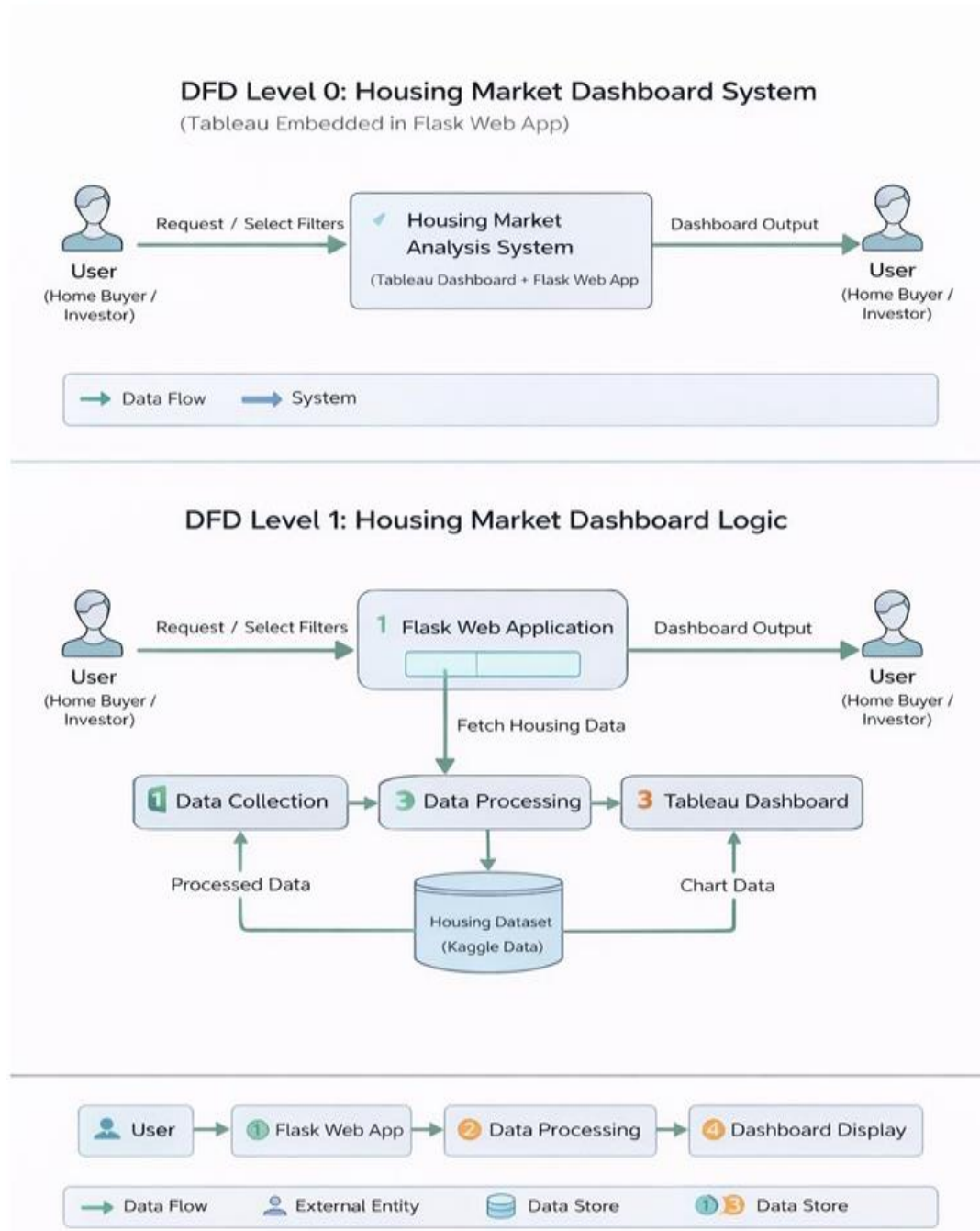
Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                                |
|--------|----------------------------|------------------------------------------------------------|
| NFR-1  | <b>Usability</b>           | Users can easily view and filter housing dashboard.        |
| NFR-2  | <b>Security</b>            | Login system protects user accounts and data.              |
| NFR-3  | <b>Reliability</b>         | Dashboard shows correct housing price data without errors. |
| NFR-4  | <b>Performance</b>         | Tableau dashboard loads fast inside Flask app.             |
| NFR-5  | <b>Availability</b>        | Web application is accessible anytime online               |
| NFR-6  | <b>Scalability</b>         | System supports more housing data and users in future.     |

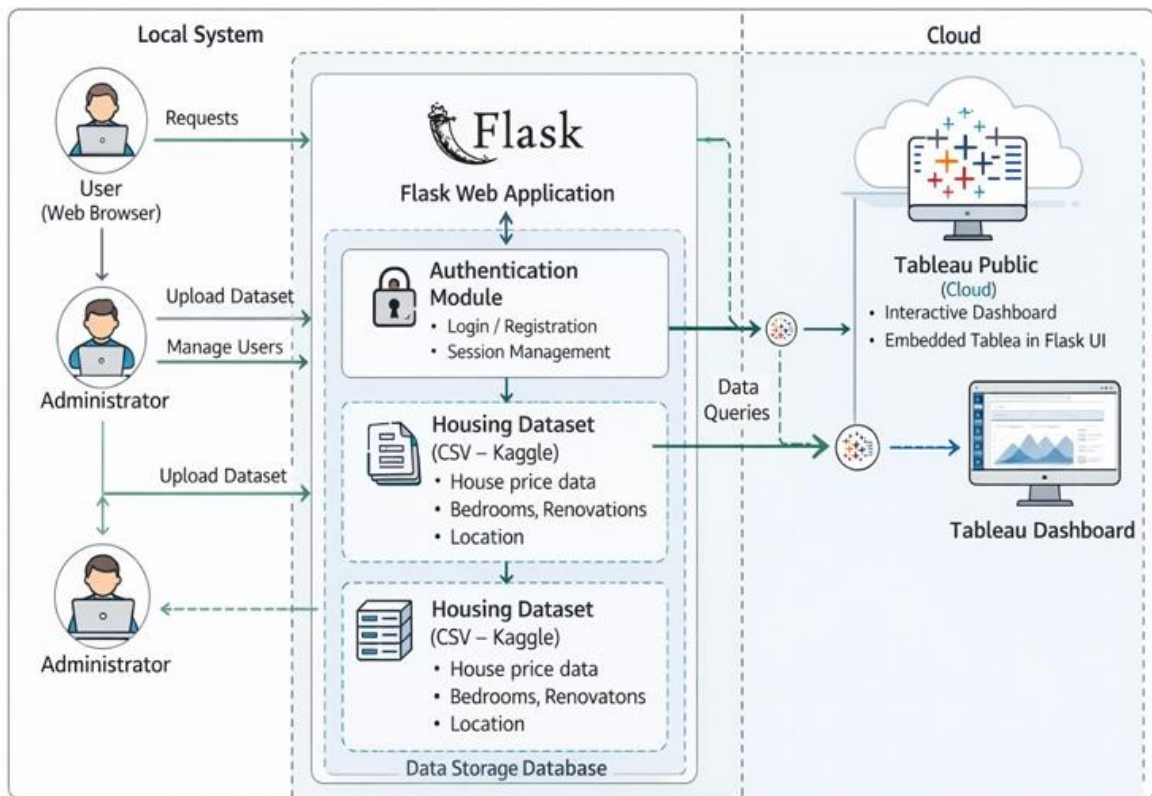
### 3.3 Data Flow Diagram

## Data Flow Diagrams:

### Flow



### 3.4 Technology Stack



| S.No | Component                       | Description                                            | Technology                            |
|------|---------------------------------|--------------------------------------------------------|---------------------------------------|
| 1    | User Interface                  | Web UI where user logs in and views dashboard          | HTML, CSS, Bootstrap                  |
| 2    | Application Logic-1             | Handles login, routing and request processing          | Python (Flask)                        |
| 3    | Application Logic-2             | Data filtering and processing logic                    | Python (Pandas)                       |
| 4    | Application Logic-3             | Embedding and displaying Tableau dashboard             | Tableau Public (Embed Code)           |
| 5    | Database                        | Stores user login details                              | SQLite                                |
| 6    | Cloud Database                  | Not used in this project                               | Not Applicable                        |
| 7    | File Storage                    | Stores housing dataset CSV file                        | Local File System                     |
| 8    | External API-1                  | Interactive dashboard visualization service            | Tableau Public                        |
| 9    | External API-2                  | Gmail login (if implemented)                           | Google OAuth API                      |
| 10   | Machine Learning Model          | Not used in this system                                | Not Applicable                        |
| 11   | Infrastructure (Server / Cloud) | Application runs locally and connects to Tableau Cloud | Local Server (Flask) + Tableau Public |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                     | Technology                       |
|------|--------------------------|-------------------------------------------------|----------------------------------|
| 1    | Open-Source Frameworks   | Backend framework used for development          | Flask (Python)                   |
| 2    | Security Implementations | Secure login, password hashing, session control | Flask Session, Werkzeug Security |
| 3    | Scalable Architecture    | Modular design allowing future data expansion   | Flask Architecture               |
| 4    | Availability             | Accessible when server is running               | Local Server / Cloud Hosting     |
| 5    | Performance              | Fast dashboard loading and efficient filtering  | Pandas + Tableau Rendering       |

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## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

Problem:

Raw housing data is difficult to interpret.

Solution:

Interactive Tableau dashboard and story embedded in Flask.

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### 4.2 Proposed Solution

The system:

- Downloads housing dataset
  - Cleans and prepares data
  - Creates 6 unique visualizations
  - Builds dashboard
  - Creates storyboard
  - Publishes to Tableau Public
  - Embeds dashboard in Flask UI
- 

### 4.3 Solution Architecture

User

↓

Flask Web Application

↓

Embedded Tableau Dashboard



↓

Tableau Public Server

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## **5. PROJECT PLANNING & SCHEDULING**

### **5.1 Project Planning**

| <b>Phase</b> | <b>Activity</b> |
|--------------|-----------------|
|--------------|-----------------|

|         |                 |
|---------|-----------------|
| Phase 1 | Data Collection |
|---------|-----------------|

|         |                  |
|---------|------------------|
| Phase 2 | Data Preparation |
|---------|------------------|

|         |                    |
|---------|--------------------|
| Phase 3 | Data Visualization |
|---------|--------------------|

|         |                    |
|---------|--------------------|
| Phase 4 | Dashboard Creation |
|---------|--------------------|

|         |                |
|---------|----------------|
| Phase 5 | Story Creation |
|---------|----------------|

|         |                     |
|---------|---------------------|
| Phase 6 | Performance Testing |
|---------|---------------------|

|         |                 |
|---------|-----------------|
| Phase 7 | Web Integration |
|---------|-----------------|

|         |               |
|---------|---------------|
| Phase 8 | Documentation |
|---------|---------------|

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## **6. FUNCTIONAL AND PERFORMANCE TESTING**

### **6.1 Performance Testing**

#### **Amount of Data Loaded**

- Housing dataset loaded successfully

#### **Utilization of Filters**

- Sale Price – Top N Filter
- Age of House – Top N Filter

#### **Number of Calculation Fields**

- House Age (Calculated Field)

#### **Number of Visualizations**

1. Count of Transformed Housing Data
2. Total Sales by Years Since Renovation
3. Distribution of House Age by Renovation Status
4. House Age Distribution by Bathrooms, Bedrooms, and Floors

Total Unique Visualizations: **4**

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## 7. RESULTS

### 7.1 Output Screenshots

(Insert the following screenshots)

- Visualization Sheets
- Dashboard
- Story
- Filters
- Flask Web UI

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## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Easy understanding of housing trends
- Interactive filters
- Web-based accessibility
- Clear data insights

### Disadvantages

- Depends on Tableau Public server
- Requires internet connection

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## 9. CONCLUSION

The project successfully visualizes housing market trends using Tableau and integrates the dashboard with Flask. The system helps users analyze sale price patterns, renovation impact, and house age distribution in an interactive way.

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## 10. FUTURE SCOPE

- Add Machine Learning prediction model
  - Real-time data integration
  - Cloud deployment
  - User authentication system
-

## 11. APPENDIX

- **Source Code:**

- ❖ **Flask code:**

```
from flask import Flask, render_template
```

```
app = Flask(__name__)
```

```
@app.route('/')
```

```
def home():
```

```
    return render_template('index.html')
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```

- ❖ **Index.html [html code]:**

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Housing Market Trends</title>
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
<style>
```

```
html { scroll-behavior: smooth; }
```

```
body {
```

```
    font-family: 'Segoe UI', sans-serif;
```

```
}
```

```
/* Navbar */
```

```
.navbar {  
  background: #0d6efd;  
}
```

```
.navbar .nav-link,  
.navbar-brand {  
  color: white !important;  
  font-weight: 500;  
}
```

```
/* Sections Animation */
```

```
section {  
  min-height: 100vh;  
  padding: 100px 0;  
  opacity: 0;  
  transform: translateY(40px);  
  transition: 0.8s ease-in-out;  
}
```

```
section.show {  
  opacity: 1;  
  transform: translateY(0);  
}
```

```
/* Home */
```

```
.home {  
  background: linear-gradient(to right, #4e73df, #1cc88a);  
  color: white;  
  text-align: center;  
}
```

```
/* Buttons */
```

```
.btn-custom {  
  padding: 10px 25px;
```

```
border-radius: 30px;
transition: 0.3s;
}
```

```
.btn-custom:hover {
  transform: scale(1.08);
}
```

```
/* Iframe */
```

```
iframe {
  border-radius: 15px;
  box-shadow: 0 8px 20px rgba(0,0,0,0.2);
  transition: 0.3s;
}
```

```
iframe:hover {
  transform: scale(1.02);
}
```

```
footer {
  background: #0d6efd;
  color: white;
  text-align: center;
  padding: 15px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- NAVBAR -->
```

```
<nav class="navbar navbar-expand-lg fixed-top">
```

```
<div class="container">
```

```
<a class="navbar-brand" href="#home"> Housing Trends</a>
```

```
<div>
```

```
<a class="nav-link d-inline" href="#home">Home</a>

<a class="nav-link d-inline" href="#dashboard">Dashboard</a>

<a class="nav-link d-inline" href="#story">Story</a>

</div>

</div>

</nav>
```

```
<!-- HOME -->

<section id="home" class="home show d-flex align-items-center">

<div class="container">

  <h1 class="display-4 fw-bold" id="title"></h1>

  <p class="lead mt-3">Interactive Tableau Dashboard & Story using Flask</p>

  <a href="#dashboard" class="btn btn-light btn-custom mt-4">

    View Dashboard ↓

  </a>

</div>

</section>
```

```
<!-- DASHBOARD -->

<section id="dashboard" class="bg-light">

<div class="container text-center">

  <h2 class="mb-4"><img alt="Tableau logo" data-bbox="264 606 281 618"/> Housing Market Dashboard</h2>

  <iframe

src="https://public.tableau.com/views/Book1_17696952321770/Dashboard1?:showVizHome=no&:embed=true"

width="100%"

height="700"

frameborder="0"

allowfullscreen>

</iframe>

<br><br>
```

```
<a href="#home" class="btn btn-primary btn-custom">Home</a>

<a href="#story" class="btn btn-success btn-custom">Next → Story</a>

</div>

</section>

<!-- STORY -->

<section id="story" class="bg-white">

<div class="container text-center">

  <h2 class="mb-4">🏠 Housing Market Story</h2>

  <iframe

    src="https://public.tableau.com/views/Book1_17696952321770/Story1?:showVizHome=no&:embed=true"

    width="100%"

    height="700"

    frameborder="0"

    allowfullscreen>

  </iframe>

  <br><br>

  <a href="#home" class="btn btn-dark btn-custom">Back to Home</a>

</div>

</section>

<footer>

© 2026 Internship Project | Tableau + Flask

</footer>

<script>

/* Scroll Animation */

const sections = document.querySelectorAll("section");

window.addEventListener("scroll", () => {

  const trigger = window.innerHeight * 0.85;
```

```
sections.forEach(sec => {  
  if(sec.getBoundingClientRect().top < trigger){  
    sec.classList.add("show");  
  }  
});  
});  
  
/* Typewriter Effect */  
const text = "Visualizing Housing Market Trends";  
let i = 0;  
function typing(){  
  if(i < text.length){  
    document.getElementById("title").innerHTML += text.charAt(i);  
    i++;  
    setTimeout(typing, 50);  
  }  
}  
typing();  
  
</script>  
  
</body>  
</html>
```



- **Dataset Link:**

- ❖ **Kaggle Housing Dataset:**

Visualizing Housing Market Trends An Analysis of Sale Prices and Features using Tableau

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

## GitHub & Project Demo Link:

- ❖ **GitHub link:**

<https://github.com/Lakshmi130604/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau/tree/main>

- ❖ **DEMO VIDEO LINK:**

[https://drive.google.com/file/d/1df7i8HB70rKN\\_WyNZiaY7N8xr1\\_j21z0/view?usp=sharing](https://drive.google.com/file/d/1df7i8HB70rKN_WyNZiaY7N8xr1_j21z0/view?usp=sharing)

- ❖ **DASHBOARD LINK:**

[https://public.tableau.com/views/Book1\\_17696952321770/Dashboard1?:showVizHome=no&embed=true](https://public.tableau.com/views/Book1_17696952321770/Dashboard1?:showVizHome=no&embed=true)

- ❖ **STORY LINK:**

[https://public.tableau.com/views/Book1\\_17696952321770/Story1?:showVizHome=no&embed=true](https://public.tableau.com/views/Book1_17696952321770/Story1?:showVizHome=no&embed=true)

- **Project Web Interface LINK:**

<http://127.0.0.1:5000>